



Politecnico di Torino

Communications Systems

Assignment 3 Report

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December 2022

In this assignment, we focus on gold codes, CDMA performance and Hartogs Hughes algorithm.

Exercise 1

At first, we want to generate gold sequences. But first let's look what a gold sequence is. We have two m-sequence c_1 and c_2 generated by two LFSR made up from m cells and two paired primitive polynomials $p_1(D)$ and $p_2(D)$ respectively. A set of gold codes is made from having these two sequences and binary sum of first sequence and a cyclic shift of second sequence, and because we have $2^m - 1$ cyclic shifts, so a set of gold codes is made from 129 sequences.

When we want to implement a gold sequence in MATLAB, there is an index which ranges from -2 to $2^m - 2$. First two indices, -2 and -1, belong to the original sequences and indices from 0 to 126 belong to remaining 127 sequences which are obtained through binary summing of c_1 and a cyclic shift of c_2 .

For the first question, we are asked to plot autocorrelation function for a gold sequence with index ≥ 0 . We know that if we plot autocorrelation function for a gold sequence with index < 0 , we will have a shape which is $N - 1$ in the zero and -1 in other places which is shown in Figure 1-1.

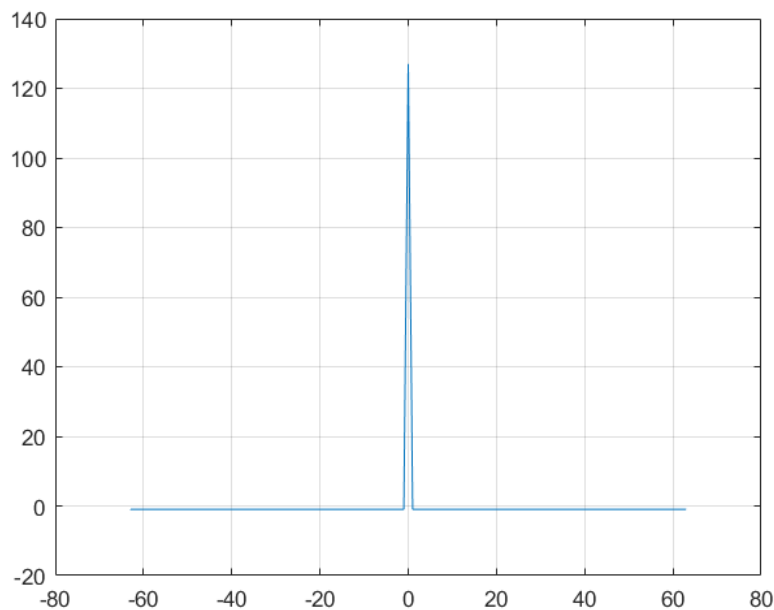


Figure 1-1: Autocorrelation function for a gold sequence with index < 0

Now we want to know what happens if we have a gold sequence with index ≥ 0 . The result is shown in Figure 1-2.

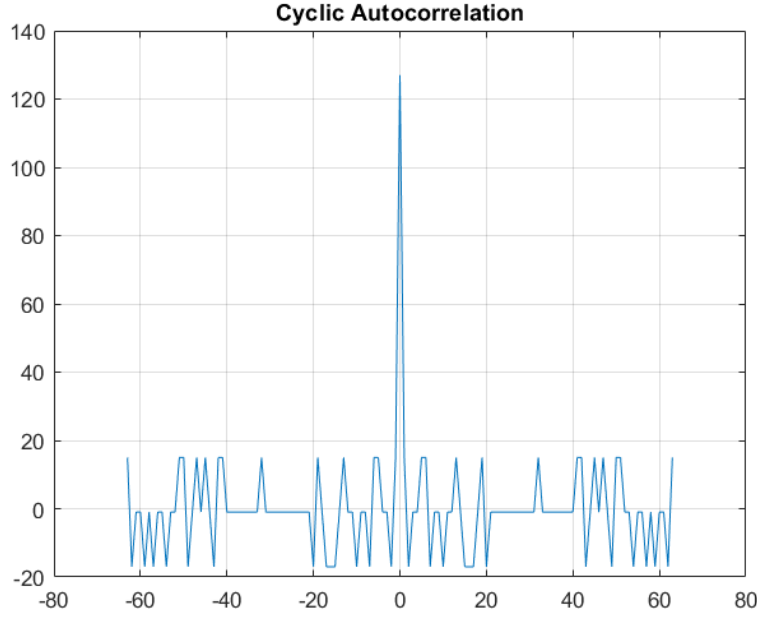


Figure 1-2: Autocorrelation function for a gold sequence with index ≥ 0

We can observe that the peak is again at $2^m - 1 = 127$ and it is in zero, but at other locations it is not -1 and it has 3 values of $-17, -1$, and -15 . We expect this because we know that in gold sequences, autocorrelation property is good but not nearly optimal as for m-sequences.

For the next part, we know that for cross correlation of two gold sequences, we have three-value property. In other words, for two gold sequences, the result of cross correlation is :

$$\begin{cases} -t \\ -1 \\ t-2 \end{cases}, t = 2^{\frac{m+1}{2}} + 1 \text{ for odd } m, t = 2^{\frac{m+2}{2}} + 1 \text{ for even } m.$$

At first, we compute and plot the cross correlation function for condition in which both indices ≥ 0 . The result is shown in Figure 1-3.

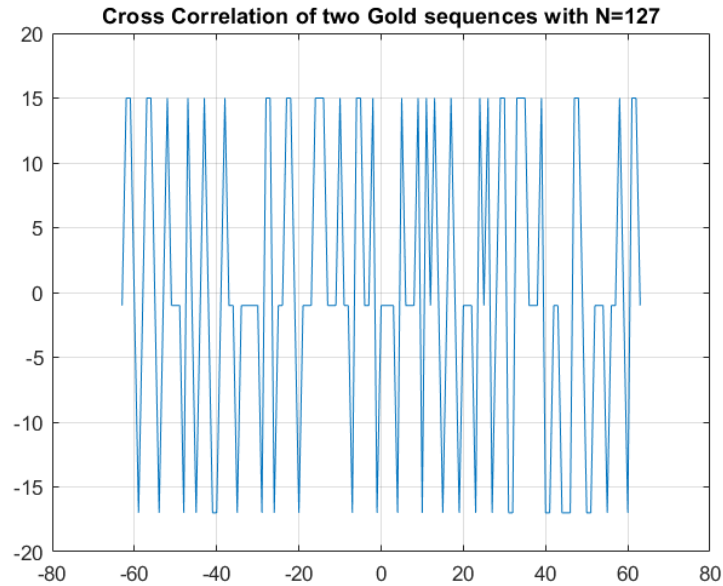


Figure 1-3: Cross correlation function for two gold sequences with index ≥ 0

It can be seen that three-value property is verified. Now we repeat this for situation in which one index is greater than zero and the other is less than zero. The result is shown in Figure 1-4.

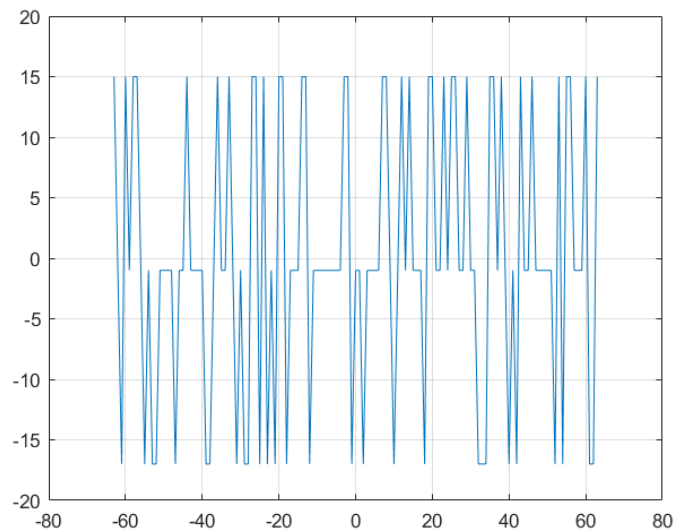


Figure 1-4: Cross correlation function for two gold sequences with one negative index

We can see that still three-value property is verified. So if one of the sequences is not an m-sequence and is a gold sequence, the three-value property is verified.

For the next part, we have to compute and plot Welch and Sidelnikov bounds which are bounds for the maximal cyclic correlation magnitude r_M . But let's look at what r_M is. For a set of K bipolar sequences with of period N we have three values:

- The maximal out-of-phase cyclic autocorrelation magnitude r_A which is computed through $r_A = \max \left\{ \left| r_{c_i'}(\tau) \right| \right\} \quad 1 \leq \tau \leq N-1, \quad 1 \leq i \leq K$. An important note is this is out-of-phase autocorrelation, meaning that maximum value which is for $\tau = 0$ is not included.
- The maximal cyclic cross correlation magnitude r_C which is computed through $r_C = \max \left\{ \left| r_{c_i' c_j'}(\tau) \right| \right\} \quad 0 \leq \tau \leq N-1, \quad 1 \leq i, j \leq K, \quad i \neq j$.
- The maximal cyclic correlation magnitude r_M which is computed through $r_M = \max \{r_A, r_C\}$

Welch bound for complex sequences of symbols with unitary magnitude is defined as:

$$r_M \geq \left[N^{2s} \sqrt{\frac{1}{KN-1} \left[\frac{KN}{\binom{N+s-1}{s}} \right]} \right], \quad s = \{1, 2, 3, \dots\}$$

This formula is simplified as:

$$r_M \geq \left[N^2 \sqrt{\frac{K-1}{KN-1}} \right]$$

And even if K is approximately large, it is simplified as:

$$r_M \geq \lceil \sqrt{N} \rceil$$

Since we are working with binary sequences, we use the ceil of the above formulas.

Sidelnikov bound holds for binary bipolar sequences and is defined as:

$$r_M \geq \left\lceil \sqrt{[(2s+1)(N-s) + \frac{s(s+1)}{2} - \frac{2^s N^{2s+1}}{K(2s)! \binom{N}{s}}]} \right\rceil, \quad s = \{1, 2, 3, \dots\}, \quad 0 \leq s \leq \frac{2N}{5}$$

The Sidelnikov bound is tighter than Welch bound.

Now we plot three versions of Welch bound and Sidelnikov bound for $N = 127$ and $1 \leq K \leq 200$. The result is shown in Figure 1-5.

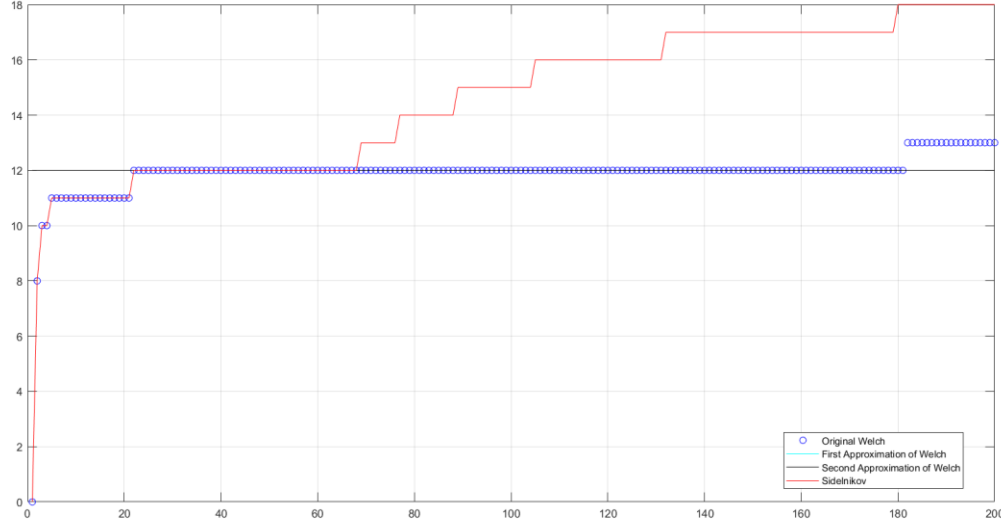


Figure 1-5: Three versions of Welch bound and Sidelnikov bound for $N = 127$ and $1 \leq K \leq 200$

Now we compute r_M for the entire set of $K = 129$ gold sequences. For this task, at first, in a for loop, we start from the gold sequence with index = -2 and compute its autocorrelation and r_A and then we compute the maximum of its cross correlation with gold sequences with indices from -1 to 126 . Then we repeat this procedure for gold sequence with index = -1 but this time, we compute the maximum of its cross correlation with gold sequences with indices from 0 to 126 because its cross correlation with gold sequence with index = -2 is computed in previous iteration. We repeat this process for all 129 sequences and in each iteration, we compare the obtained r_A and r_C with current r_A and r_C and if needed, update them, and then compare the updated r_A and r_C to decide for r_M .

At the end, we add the obtained r_M to the current plot and the result is shown in Figure 1-6.

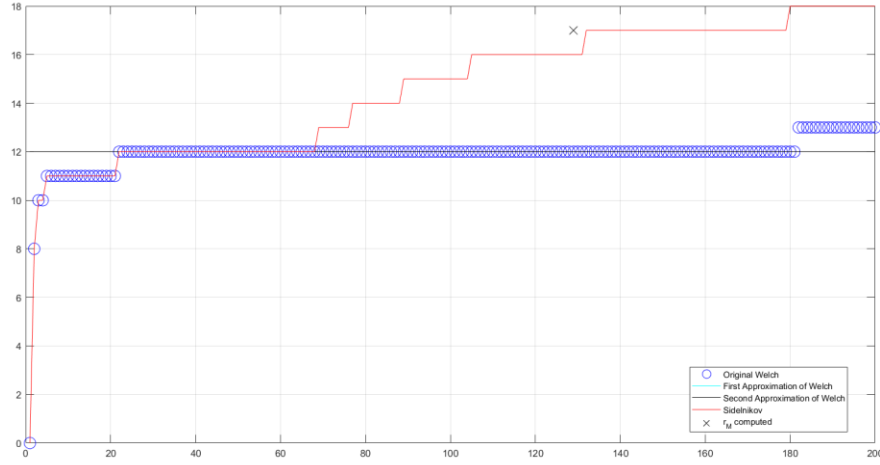


Figure 1-6: Three versions of Welch bound and Sidelnikov bound for $N = 127$ and $1 \leq K \leq 200$ and computed r_M

We can see that the gold codes performance is very close to Sidelnikov bound which is the tighter bound and therefore it shows that the performance of gold codes is very good and optimal.

For the next part, we truncate the sequences to have $N = 126$. Then we repeat the same process we did for the previous part to compute the new r_M for the truncated sequences. Then we compute the Welch and Sidelnikov bound for $N = 126$ and $1 \leq K \leq 200$. The result is shown in Figure 1-7.

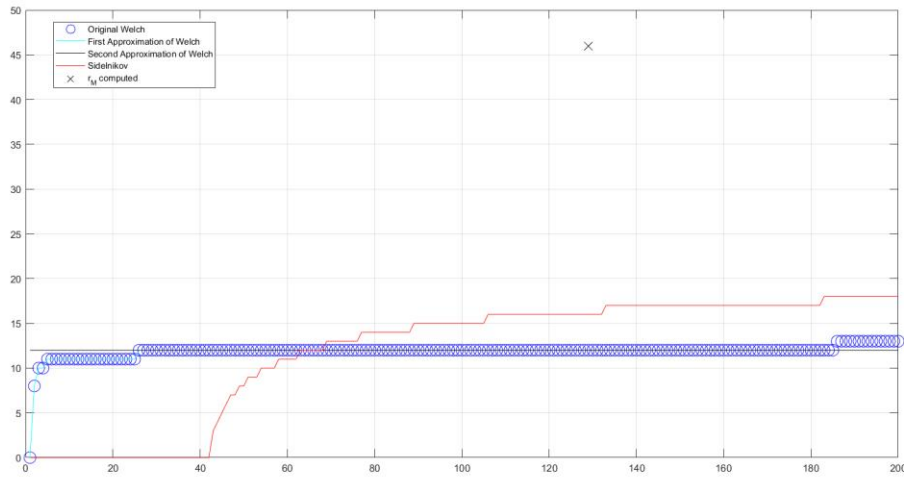


Figure 1-7: Three versions of Welch bound and Sidelnikov bound for $N = 126$ and $1 \leq K \leq 200$ and computed r_M

It is shown that the performance of truncated gold sequences is very poor in comparison with the original one and it is very far from Sidelnikov bound.

Exercise 2

In this exercise, we want to discuss CDMA performance.

In the first scenario, we have only one user. This user U_1 transmits a bit which we call v_1 and we transform it to bipolar symbol $v'_1 = 2 \times v_1 - 1$. Then this user uses c'_1 as spreading sequence. The channel adds a noise N with zero mean and variance equal to $\frac{N_0}{2}$. The resulting vector called r which is received by receiver. The receiver projects r over c'_1 by bitwise multiplication and summing all of them. If the result of sum is greater than zero, the receiver decides for 1 and if it is negative the receiver decides for 0. Then this is compared against v_1 to see if we have error or not.

At first, we compute the real values of $\frac{E_b}{N_0}$ and from this real value, we obtain the σ we need for simulating the white noise. Then in a while loop, we define v_1, v'_1, c'_1 then we project r over c'_1 and sum the result and decide that bit was 0 or 1. Then we compare it against the original bit and if they were not equal, we increase the counter for the errors. At the end, we compute the ratio of this counter and total number of iterations of while loop to obtain the error rate for that $\frac{E_b}{N_0}$.

At first, we plot the analytical bit error rate which is equal to $\frac{1}{2} \operatorname{erfc} \sqrt{\frac{E_b}{N_0}}$ which is shown in Figure 2-1.

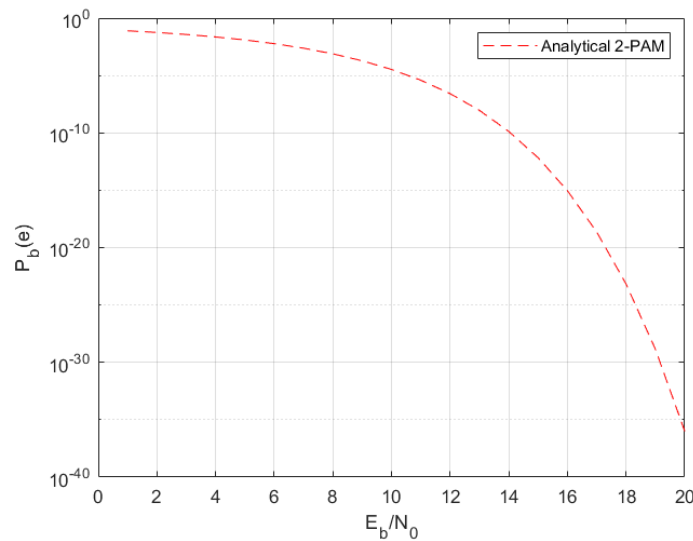


Figure 2-1: Analytical formula for error rate in one user scenario

Now we add the simulated error rate and show it in Figure 2-2.

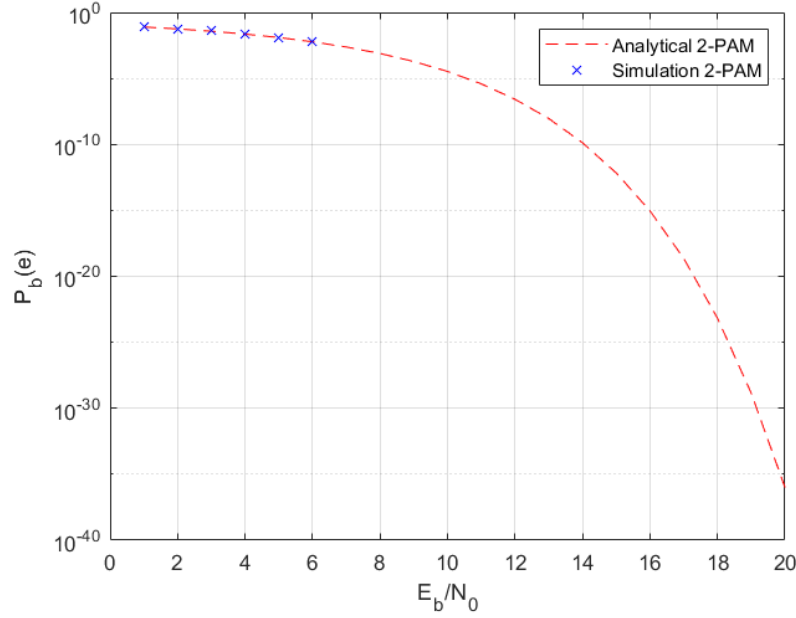


Figure 2-2: Analytical formula and simulation for error rate in one user scenario

Now we move to the second scenario. The second user transmits like the first user, and we again project the r over c'_1 which in this scenario, has an additional term for interference which is the projection of c'_2 over c'_1 . Because of this term, the performance is worsened, and the error rate is higher. We verify this in Figure 2-3.

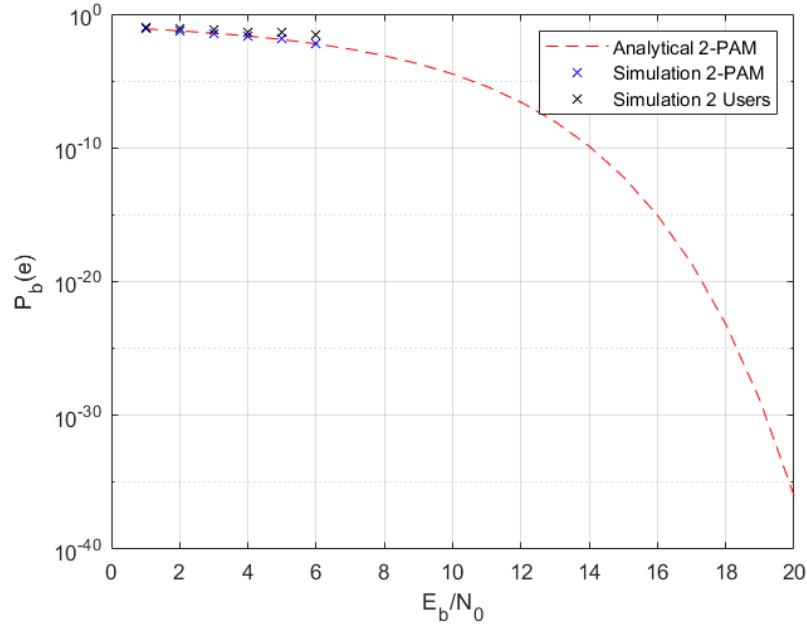


Figure 2-3: Result of one user scenario and the simulated error rate for two user scenario

For the next part, I believe that the projection of c'_2 over c'_1 is the dominating parameter for bit error rate for this scenario because it has a significant effect on r . Now we want to discuss why this happens. For example, if the projection is 0.5, half of the time, we receive -1.5 we decide for 0 but if the original bit was 1, the probability of error is obtained by $P(\text{Noise} > 1.5)$. This is a positive interference because by the shift we have from -1 to -1.5 , the error rate is decreased. But in another way, half of the time, we have the negative inference. If we receive -0.5 we still decide for 0 but now the error rate is larger and is equal to $P(\text{Noise} > 0.5)$. The combined error rate for this two-user scenario is obtained through: $P(e) = \frac{P(\text{Noise} > 0.5) + P(\text{Noise} > 1.5)}{2}$. The general equation for probability of bit error rate in two-user scenario is:

$$P(e) = \frac{\left(\frac{1}{2} \text{erfc} \sqrt{\frac{E_b}{N_0} \times (E_b + \text{Projection})^2} \right) + \left(\frac{1}{2} \text{erfc} \sqrt{\frac{E_b}{N_0} \times (E_b - \text{Projection})^2} \right)}{2}$$

Note that in this formula, we consider $E_b = 1$, $\text{Projection} = \frac{\sum_{i=1}^n c'_{1i} \cdot c'_{2i}}{N}$

In the following, we have some examples which are shown in Figure 2-4 to Figure 2-9, respectively.

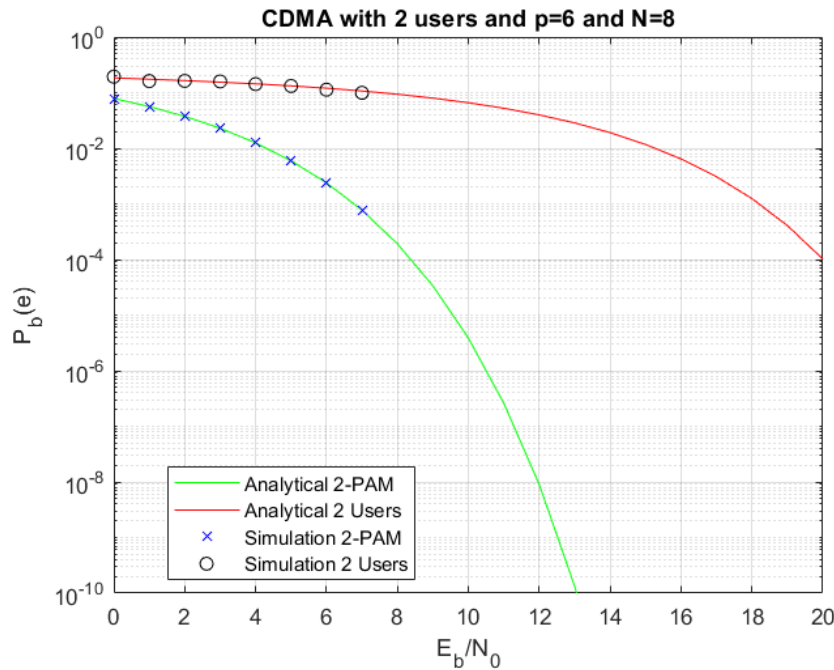


Figure 2-4: Result of one user scenario and two user scenario, projection=6, $c'_1 = 01111111$, $c'_2 = 01110111$

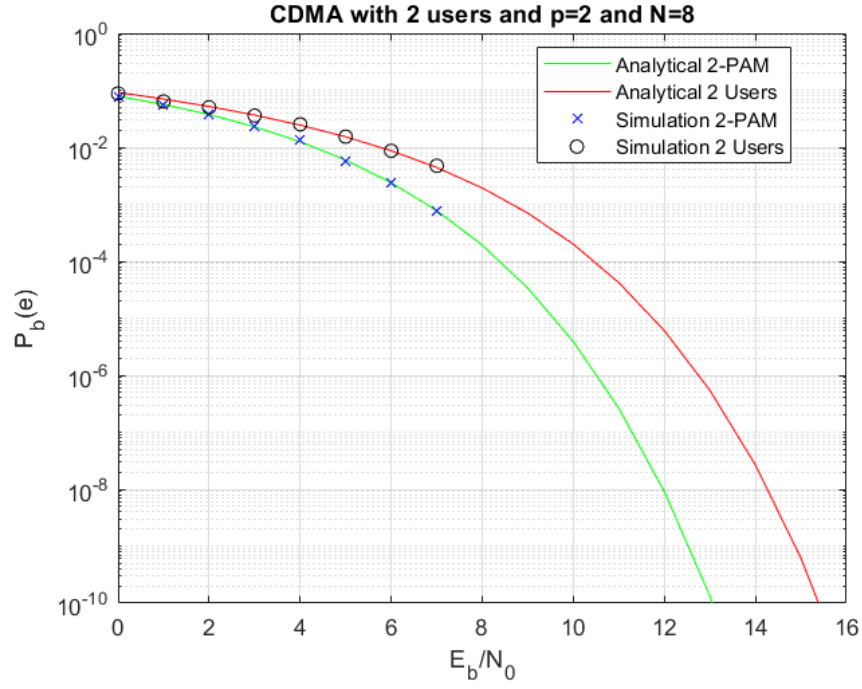


Figure 2-5: Result of one user scenario and two user scenario, projection=2,
 $c'_1 = 0110001, c'_2 = 11010001$

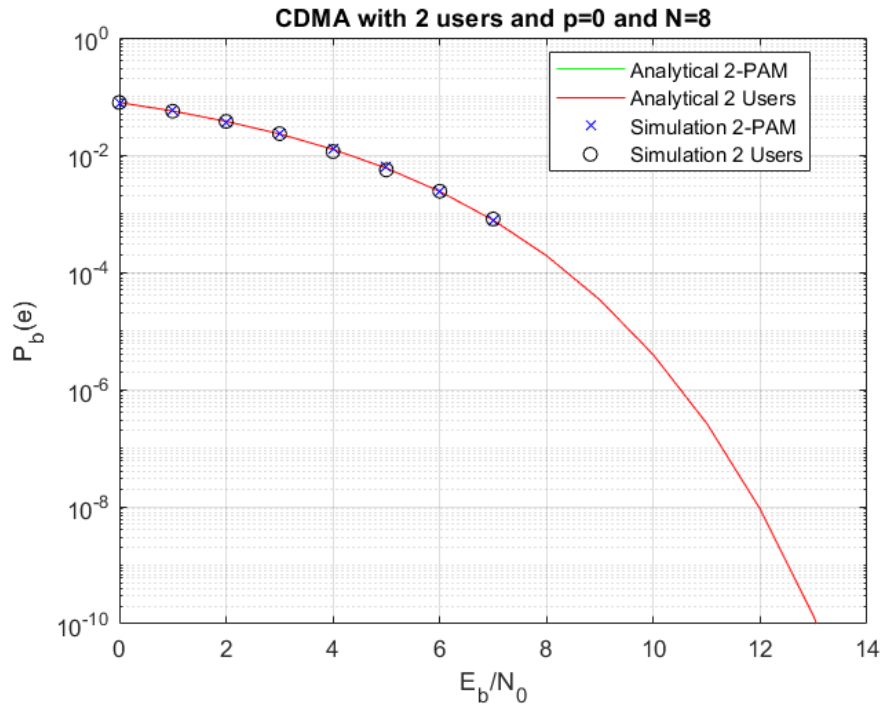


Figure 2-6: Result of one user scenario and two user scenario, projection=0,
 $c'_1 = 01000011, c'_2 = 10101011$

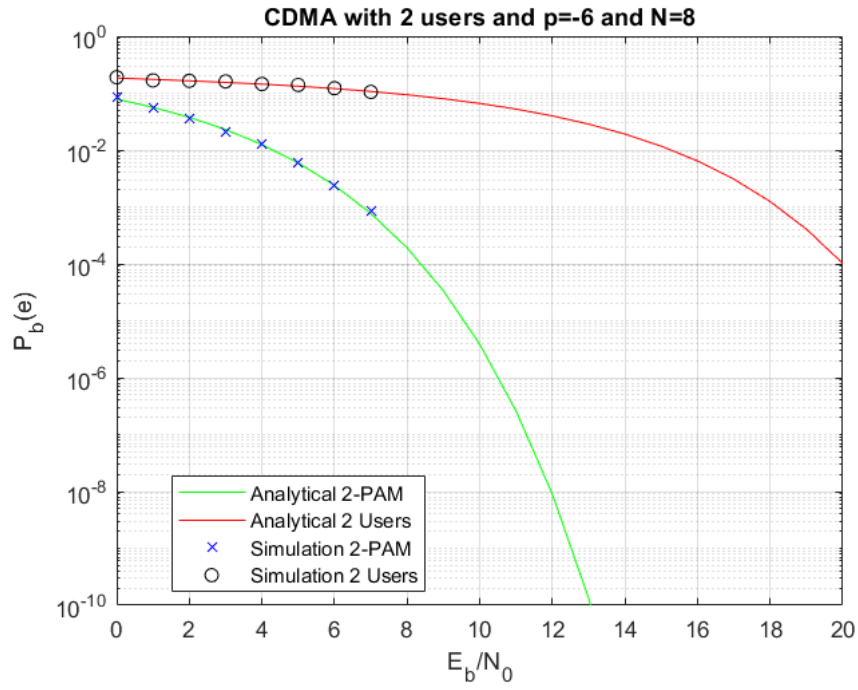


Figure 2-7: Result of one user scenario and two user scenario, projection=-6,
 $c'_1 = 11001110, c'_2 = 00110011$

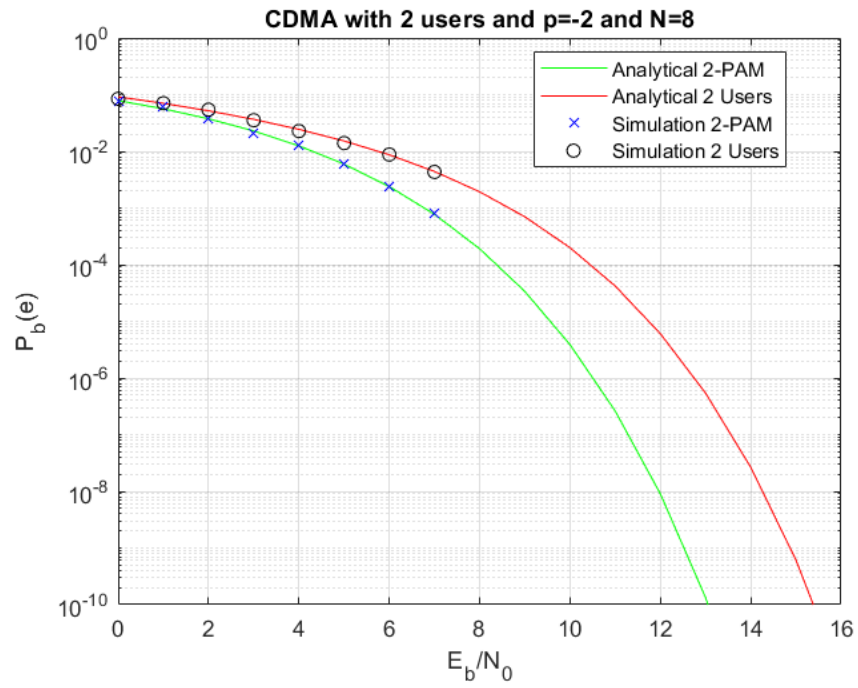


Figure 2-8: Result of one user scenario and two user scenario, projection=-2,
 $c'_1 = 11110100, c'_2 = 00010010$

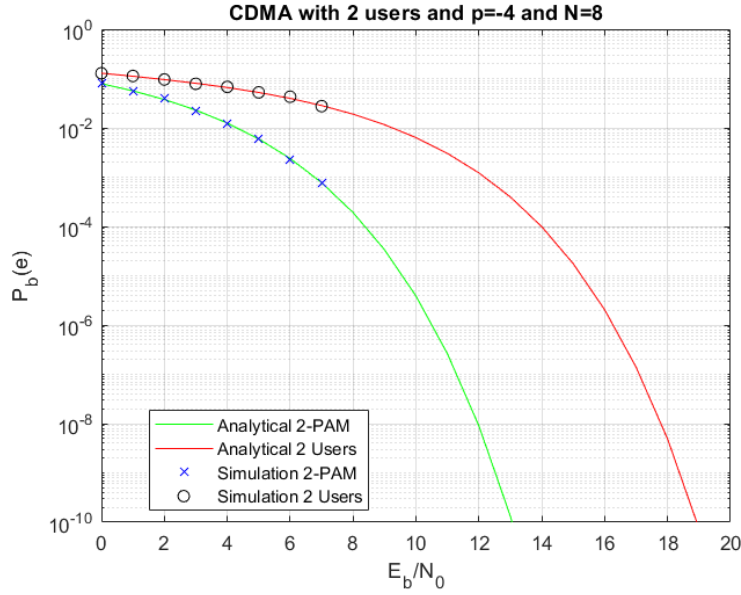


Figure 2-9: Result of one user scenario and two user scenario, projection=-4,
 $c'_1 = 00010101, c'_2 = 11001110$

We can see that simulation and analytic curve for this two-user scenario match each other perfectly. It can be seen that bit error rate for two-user scenario is significantly higher than the bit error rate for single user scenario. As we previously mentioned, this difference is because of the effect of the projection of c'_2 over c'_1 which generally increases the bit error rate. We can check this effect with equations:

When we have a single user, we send $r(t) = \pm c'_1(t) + N(t)$ and when we compute the projection of $r(t)$ over $c'_1(t)$ we have:

$$\int r(t)c'_1(t)dt = \int \pm c'_1(t)c'_1(t)dt + \int N(t)c'_1(t)dt = \pm 1 + N(0)$$

Let's look at what happens when we have two users:

$$r(t) = \pm c'_1(t) \pm c'_2(t) + N(t) \Rightarrow \int r(t)c'_1(t)dt = \int \pm c'_1(t)c'_1(t)dt + \int \pm c'_2(t)c'_1(t)dt + \int N(t)c'_1(t)dt$$

First and third terms, are same as the terms of single user scenario, but the second term which is the projection can cause positive or negative interference. Half of the time we have positive interference and half of the time we have negative interference. We call their bit error rate $p_{b1}(e)$ and $p_{b2}(e)$, respectively and we claim that we have this inequality : $\frac{p_{b1}(e)+p_{b2}(e)}{2} > p_b(e)$, where $p_b(e)$ is the bit error rate for single user scenario. We identify this with inequality with the above shown figures.

Now we move to two-user scenario with cyclic shift. In this case, at first, we choose two spreading codes, and then fix one of them and shift the other and therefore we have 8 different situations. For each SNR, we compute the bit error rate for each shift and then we average them to obtain the bit error rate for that specific SNR. Figure 2-10 to Figure 2-12 show the results.

I believe that the dominating parameter in this case is the mean of absolute values of projections for different shifts. In another words, for each shift we compute the projection, then we compute the absolute value of this projection and then average them. We compare this average against the absolute value of original projection. If the average is greater, then the performance of shifted scenario is better like Figure 2- and if they are equal, both scenarios have same bit error rate and if the average is less than the absolute value of the original projection, the not-shifted scenario performs better.

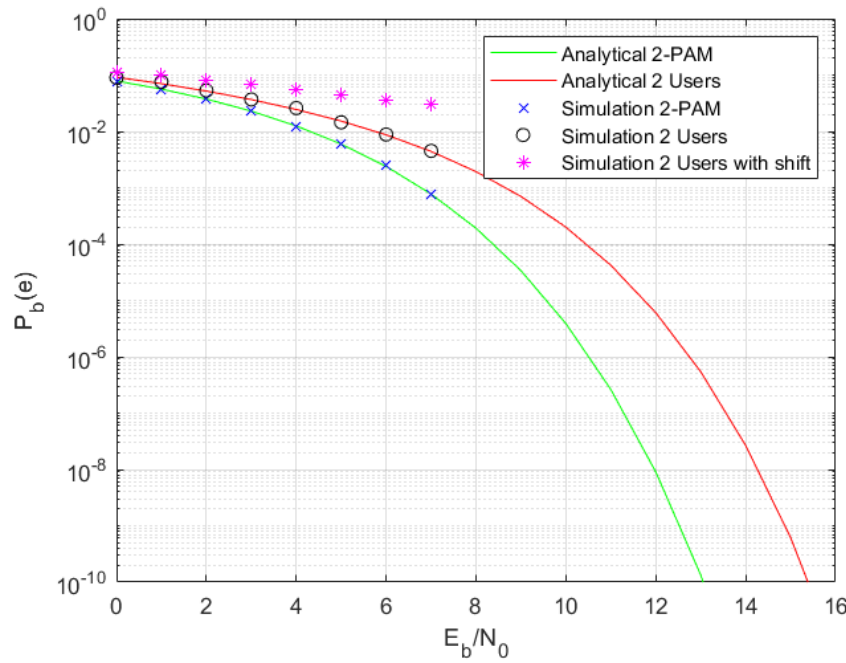


Figure 2-10: Result of one user scenario, two user scenario without cyclic shift, and two user scenario with cyclic shift with original projection=-2 and average projection=3

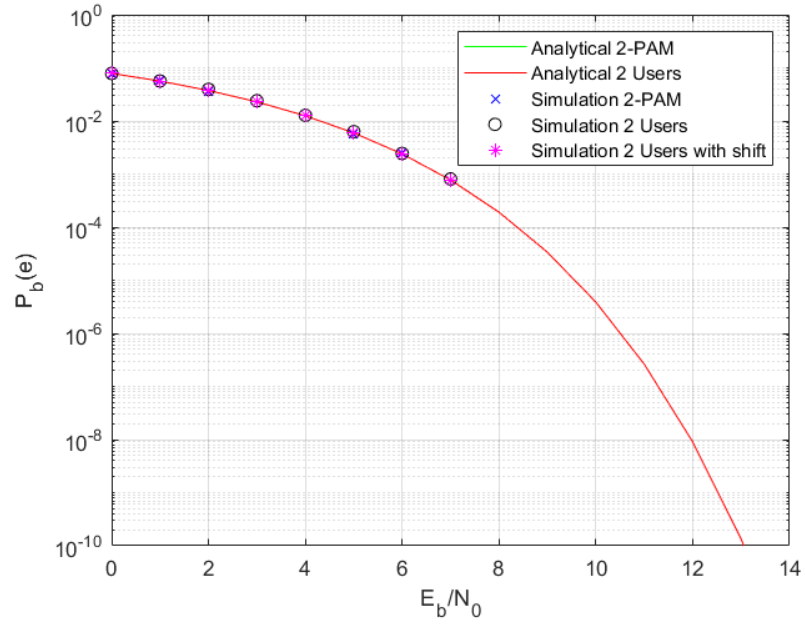


Figure 2-11: Result of one user scenario, two user scenario without cyclic shift, and two user scenario with cyclic shift with original projection=0 and average projection=0

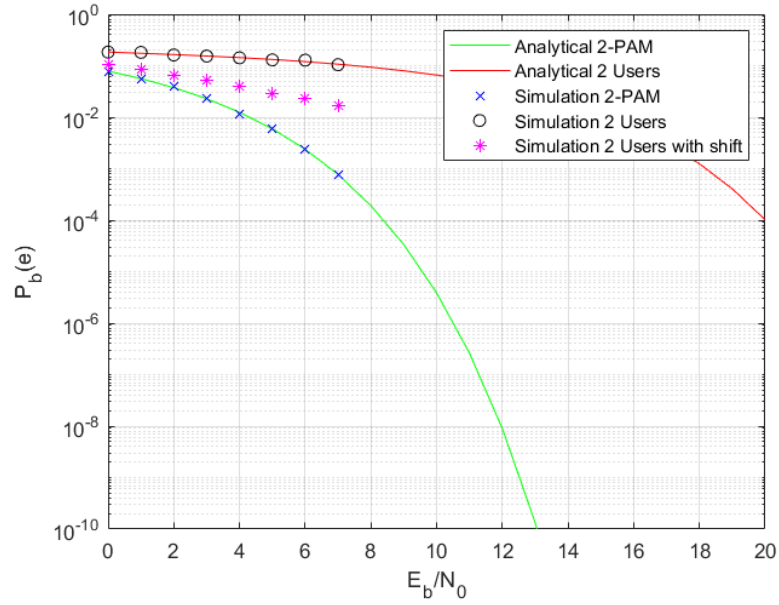


Figure 2-12: Result of one user scenario, two user scenario without cyclic shift, and two user scenario with cyclic shift with original projection=6 and average projection=2.5

Exercise 3

In this exercise, we focus on OFDM system. This system has an impulse response of $h(t) = a_1\delta(t - D_1) + a_2\delta(t - D_2) + a_3\delta(t - D_3) + a_4\delta(t - D_4)$. The values of a_i and D_i are given.

We compute $H(f)$ and focus on values $H_i = H(f = f_i)$. We compute the values of attenuation with the formula $A_i = \frac{1}{|H_i|^2}$. Then we convert this value to dB to have our attenuation profile in dB for the first part of exercise.

Then we apply the Hartogs-Hughes algorithm. Now we want to discuss how this algorithm works. Each tone in our system has an attenuation α which we name as price, and we have a total power P_{Tx} from which we “buy” bits for our tones. By buying we mean adding a bit for that tone and subtracting the attenuation from the total power. This procedure repeats until there is no remaining power. This is for the second part.

For the third part, we distribute P_{Tx} over active tones in previous part, and we can see that for same amount of active tones, we have less transmitted bits because we have a limited and decreased power dedicated for each tone. This solution is not optimal but in the case that we have many tones it is good. Because instead of working with thousands of tones that many of them have high attenuation and will be inactive, we work only with good tones that have low attenuation and in the cases when we have only microseconds to run the algorithm, this case is better.

For the last part, we distribute P_{Tx} over all active and inactive tones, and we expect to have less bits and fewer active tones because the average power for each tone has decreased in compare with previous part. The result for all parts is shown in Figure 3-1.

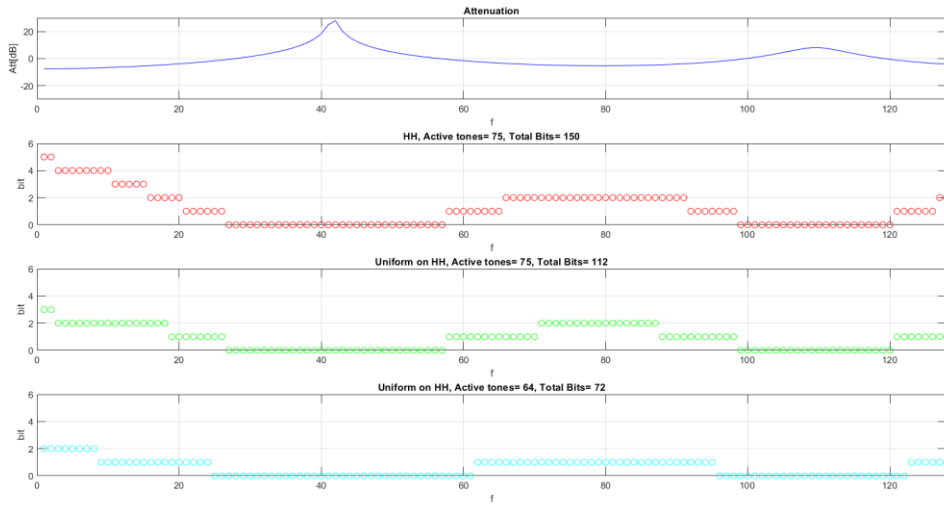


Figure 3-1: Attenuation profile, Hartogs Hughes algorithm, uniform Hartogs Hughes algorithm for active tones, Hartogs Hughes algorithm for all tones

In the following, some other examples are given in Figure 3-2 to Figure 3-6, respectively.

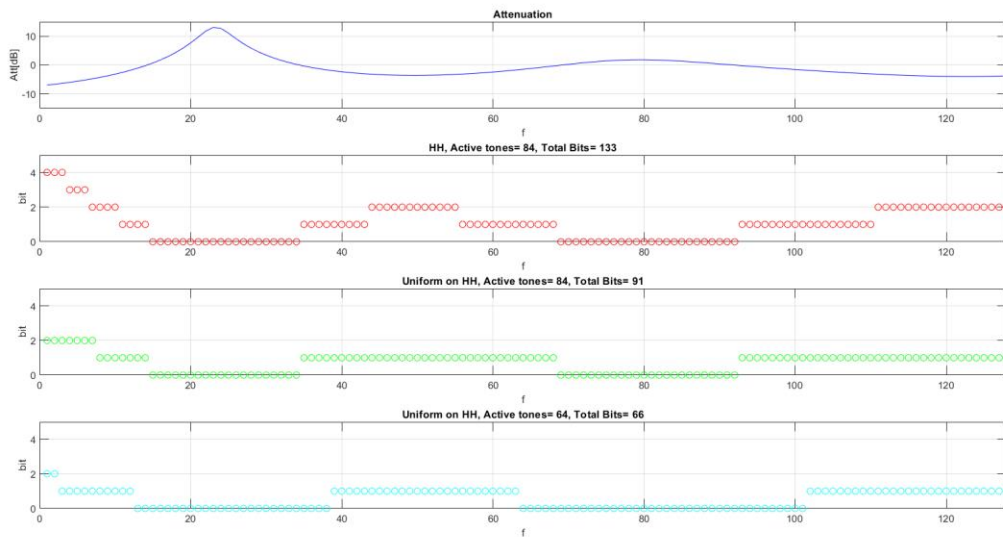


Figure 3-2: Attenuation profile, Hartogs Hughes algorithm, uniform Hartogs Hughes algorithm for active tones, Hartogs Hughes algorithm for all tones

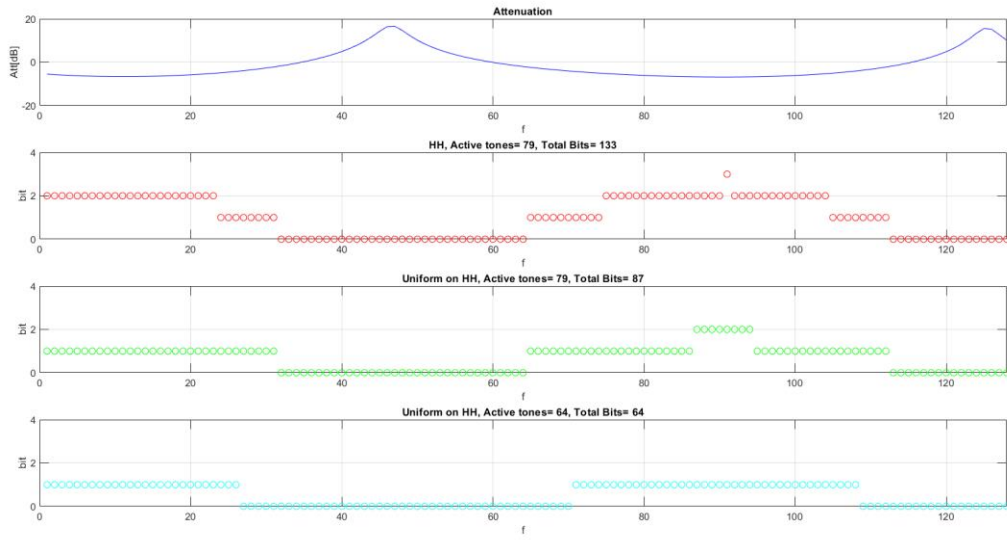


Figure 3-3: Attenuation profile, Hartogs Hughes algorithm, uniform Hartogs Hughes algorithm for active tones, Hartogs Hughes algorithm for all tones

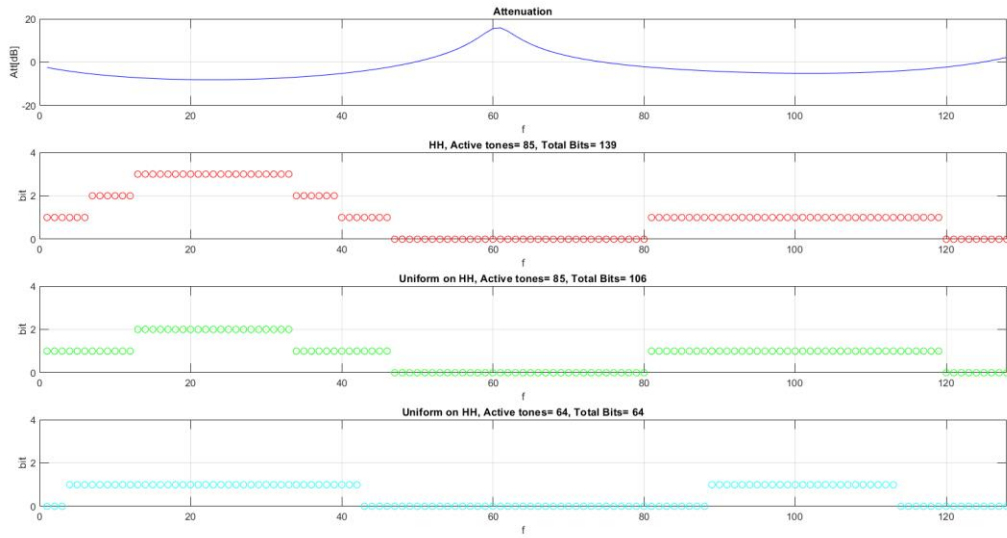


Figure 3-4: Attenuation profile, Hartogs Hughes algorithm, uniform Hartogs Hughes algorithm for active tones, Hartogs Hughes algorithm for all tones

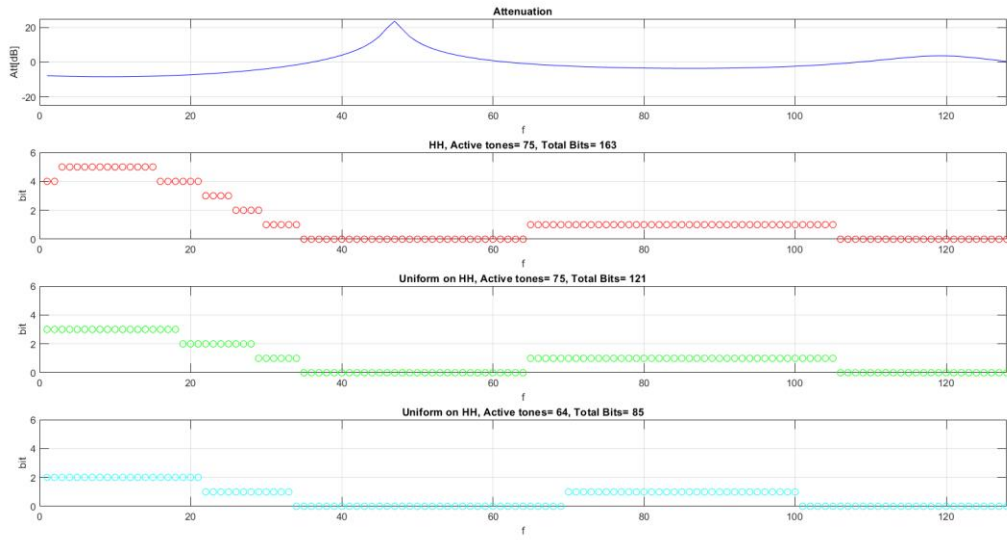


Figure 3-5: Attenuation profile, Hartogs Hughes algorithm, uniform Hartogs Hughes algorithm for active tones, Hartogs Hughes algorithm for all tones

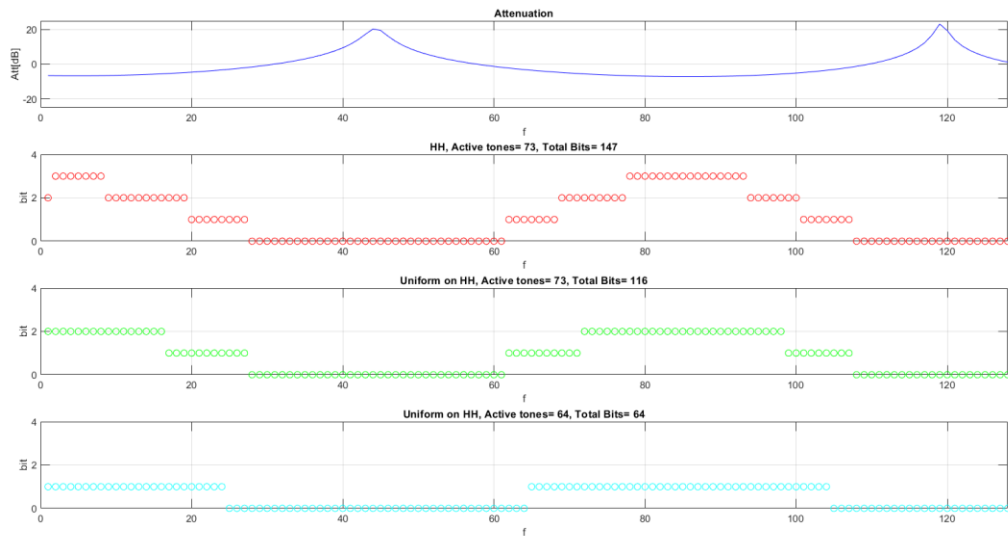


Figure 3-6: Attenuation profile, Hartogs Hughes algorithm, uniform Hartogs Hughes algorithm for active tones, Hartogs Hughes algorithm for all tones